Aman Chaudhary
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Education

Degree	Institution	CPI/%	Year
B. Tech	IIT Gandhinagar	7.77/10	2020 - Present
Class XII	Boston Public School, Agra	93%	2019-2020
Class X	D.E.I. R.E.I. Inter College, Agra	9.9/10	2017-2018

Course Projects and Internships

- **India Drought Monitor, WebGIS Developer** ***[www.indiadroughtmonitor.in]*** **[Oct'22-Dec'22]**
  (Prof. Vimal Mishra, IIT Gandhinagar)
  - Developed India Drought Monitor, a real-time drought monitoring and forecasting web application.
  - Implemented various GIS libraries and tools to represent the geospatial data on interactive and user-friendly maps.
  - The application provides valuable insights into drought conditions in India and updates on a weekly basis.
- **Geoportal of Indian Dams (GeoID), WebGIS Developer** ***[Link]*** **[May'22-July'22]**
  (Water & Climate Lab, IIT Gandhinagar)
  - Developed Geoportal of Indian Dams (GeoID), a web-based GIS application that provides information on over 5400 dams in India.
  - Utilized cutting-edge technologies such as Leaflet, QGIS, Google Earth Engine to create a user-friendly and interactive platform.
  - Implemented features such as catchment characteristics, LULC analysis using Google Earth Engine, and flood risk assessment, to provide valuable insights and data of Indian dams to users.
- **Sustainability: Energy Footprint Analysis of Food Waste** *(Prof. Udit Bhatia)* **[Jan'22-April'22]**
  - Formed a new footprint matrix to analyse the food wastage in our college mess by calculating water, carbon, energy and labour footprint of wasted food.
  - Build a website that provides a detailed analysis of daily food wastage in terms of footprints.
- ***Analysis of traffic flow models using numerical methods*** *(Prof. Satyajit Pramanik)* **[April'22]**
  - The study utilizes the OVM (Optimal Velocity Model) to predict traffic flow by finding the optimal velocity of a vehicle based on its speed and position relative to other vehicles.
  - The system is described using a second-order ODE and solved with the 4th order Runge-Kutta method, with initial conditions defined by the user to describe the vehicle's starting state. The study only considers traffic scenarios with one or two cars.

Projects

- **Web Developer, Sports IIT Gandhinagar** **[Nov'22-Present]**
  - Developed the official sports website for the Indian Institute of Technology Gandhinagar (IITGN).
  - Collaborated with the sports committee to understand the requirements and designed a user-friendly website that provides easy access to sports events, schedules, results, and achievements of the institute.
  - Utilized HTML, CSS, JavaScript, and ReactJS to build the website and ensured the site's compatibility with all modern browsers.
- **Web Developer, SynOCE IIT Gandhinagar** **[Aug'22-Oct'22]**
  - Developed the webpage for the Civil Engineer Society of Indian Institute of Technology Gandhinagar (IITGN).
  - Worked closely with the society members to understand their requirements and designed a website that accurately reflects the goals and activities of the society.
  - Implemented various features such as a member directory, event calendar, and blogs to provide a comprehensive view of the society's operations.

Skill Summary

- **Programming Languages:** JavaScript, Python (Basics), C/C++
- **Web Development:** HTML, CSS, Bootstrap, ReactJS, NodeJS, Git, GitHub, Firebase
- **Mapping Libraries:** Leaflet, OpenLayers, ArcGIS API for JavaScript, ArcGIS StoryMaps, ArcGIS Online
- **Softwares:** QGIS, ArcGIS, Google Earth Engine, SAP2000, Autodesk Inventor, Fusion 360, Excel, MATLAB, Filmora, Figma,

Relevant Courses

- Geospatial Engineering, Water Resource Engineering, Structural Analysis, Soil Mechanics, Introduction to Earth Science, Sustainability and Environment, Mechanics of Solids, Fluid Mechanics, Engineering Graphics.
- **Ongoing:** - Networks and Complex Systems, Data Structures and Algorithm, Concrete Design, Masonry Design, Steel Design, Field Survey Project, Going Places with Spatial Analysis (MOOC)

Certifications

- Spatial Data Science: The New Frontier In Analytics
- QGIS 3: from Beginner to Advanced, Udemy
- Responsive Website Development and Design Specialization, Coursera

Positions of Responsibility & Extra-Curricular Activities

- **Member, SynOCE (*Syndicate of Civil Engineers, IIT Gandhinagar*)** **[Feb'21- Present]**
  - Technical in-charge of the society. Developed and regularly update the SynOCE website.
  - Developed new tools which can assist the team in key responsibilities.
- **Member, Mean Mechanics *(Robotics club, IIT Gandhinagar)*** **[Feb'21- Present]**
  - Conducted the workshops on Home automation and Arduino basics
- **Event Organiser, Hallabol *(Annual Intra-College sports fest)*** **[March'21-June'22]**
  - Organized various intra college sports events along with other team members
- **Event Coordinator, ATHE-League** **[March'21 -June'22]**
  - Organized Athletics league for IITGN Community.
- **Member, Sports Council** **[March'21-June'22]**
- **Council Member, MAPRC *(Media and Public Relations Cell, IIT Gandhinagar)*** **[Oct'21-July'22]**
- **Marketing Executive, Amalthea'20 *(Annual Technical Summit, IIT Gandhinagar)*** **[Nov'20-April'21]**
- **Member, Inter-IIT *Weightlifting* Contingent (20*22*)** **[Dec'22]**